Junli Gong

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Research Interests

Efficient machine learning systems—LLM inference, GPU kernels, and heterogeneous accelerators—and multimodal models that perceive, reason, and act.

Education

Northeastern University Sep 2025 – Dec 2026 (expected)
Master of Science in Computer Science, Khoury College of Computer Sciences *Seattle, WA, USA*

- **GPA:** 3.89/4.00; **Seattle Campus Scholarship** (merit-based)
- **Relevant Courses:** Machine Learning, Scalable Distributed Systems, Reinforcement Learning, Algorithms, Foundations of AI, Practical AI

Beijing Normal-Hong Kong Baptist University (BNBU) Sep 2021 – Jun 2025
Bachelor of Science (Honours) in Computer Science and Technology *Zhuhai, China*

- **Honors:** President's Honour Roll; Dean's List (4 semesters)
- **Relevant Courses:** Neural Networks and Deep Learning, Machine Learning, Data Structures and Algorithms, Operating Systems, Database Management Systems

Publications

* denotes equal contribution.

Peer-reviewed

[1] Guo, Y.*, **Gong, J.***, Cai, H., Cheung, Y.-m., & Su, W. (2026). Can Segmentation Models Understand the World? Proactive Affordance Reasoning via Vision-Grounded Chain-of-Thought. *Proceedings of EMNLP 2026* (Main Conference).

[2] Guo, Y., **Gong, J.**, Wang, W., Cai, H., Cheung, Y.-m., & Su, W. (2026). PEAM: Parametric Embodied Agent Memory for Continual Skill Learning via Internalization of Experience in Minecraft. *Findings of EMNLP 2026*.

[3] Guo, Y., **Gong, J.**, Dong, W., Cheung, Y.-m., & Su, W. (2026). Adding Thermal Awareness to Visual Systems in Real-Time via Distilled Diffusion Models. *CVPR 2026 Workshop on World Models Meet Active Sensing (WMAS)*; extended version under review at NeurIPS 2026.

[4] Zheng, C.*, **Gong, J.***, Guo, J., Tang, Z., & Wang, T. (2025). Enhancing Federated Learning in IoT Through Dynamic Spectral Clustering. *WASA 2025*.

[5] Fei, C., Xia, N., Tsai, P.-W., Lu, Y., Pan, X., & **Gong, J.** (2024). An Effective Feature Selection Algorithm for Machine Learning-based Malicious Traffic Detection. *AsiaJCIS 2024*.

Under Review

[6] Guo, Y.*, **Gong, J.***, Lu, Y., Xu, X., Cheung, Y.-m., & Su, W. (2026). Bringing Multimodal Large Language Models to Infrared-Visible Image Fusion Quality Assessment. Under review at *NeurIPS 2026*. arXiv:2605.06969.

[7] Guo, Y., **Gong, J.**, Cheung, Y.-m., & Su, W. (2026). LumiVideo: An Intelligent Agentic System for Video Color Grading. Under review at *AAAI 2027*. arXiv:2604.02409.

[8] Guo, Y., **Gong, J.**, Cheung, Y.-m., & Su, W. (2026). Fuse4Seg: Image Fusion for Multi-Modal Medical Segmentation via Bi-level Optimization. Under review at *AAAI 2027*. arXiv:2409.10328.

Research Experience

Graduate Researcher Jan 2026 – Present
Khoury College of Computer Sciences, Northeastern University *Advisor: Prof. Yifan Hu*

- Investigating speculative expert pre-dispatch for Mixture-of-Experts inference: lightweight linear probes predict expert routing from pre-attention hidden states so that the expert-parallel AllToAll can overlap with attention, while a verify-and-fallback step keeps outputs identical to the serial path.
- Built a prototype in SGLang and evaluated it on DeepSeek-V2-Lite on 2×A100; currently refining the routing predictor and the evaluation methodology.

Research Assistant Sep 2025 – Present
Department of Computer Science, Beijing Normal-Hong Kong Baptist University (remote) *Supervisor: Prof. Weifeng Su*

- Responsible for the experimental work of six papers on multimodal perception and embodied AI—implementing the methods, running training and evaluation, and writing the experiments sections; two accepted at EMNLP 2026 (Main and Findings), one at a CVPR 2026 workshop, and three under review at NeurIPS 2026 and AAAI 2027 [1–3, 6–8].
- Co-first-authored SegWorld [1] (EMNLP 2026): implemented the two-pass reasoning pipeline (instruction-free proactive observation, then instruction-conditioned visual chain-of-thought) on Sa2VA-Qwen2.5-VL-7B and evaluated it on Intent2Part, the paper's intent-to-part benchmark derived from InstructPart, where it substantially outperforms fine-tuned Sa2VA and LISA on intent-level instructions.

- Co-first-authored FuScore [6]: built the MLLM-based infrared–visible fusion quality assessor, from zero-shot scoring to fine-tuning with soft-label distributional supervision and Thurstone-based ranking objectives, reaching state-of-the-art correlation with human preference.
- Implemented and evaluated the parametric agent-memory framework (MoE-LoRA adapters, failure–correction contrastive training) in Minecraft [2] and the distilled-diffusion real-time thermal–visible fusion module with closed-loop CARLA driving evaluation [3].

Research Intern Jul 2025 – Aug 2025
Pazhou Lab / School of Computer Science and Engineering, South China University of Technology *Supervisor: Prof. Lu Lu*

- Developed a custom FP16 Tensor Core HGEMM kernel in CUDA for Transformer workloads on NVIDIA A800, combining coalesced global-memory access ($\sim 10\times$ gain over the strided-access version) with double-buffered software pipelining (+6–9%).
- Benchmarked against cuBLAS on five BERT/LLaMA-derived matrix shapes; outperformed cuBLAS by $1.33\times$ on the bandwidth-bound attention-score shape ($1024 \times 1024 \times 64$) and produced a bottleneck analysis guiding further optimization.
- Systematically evaluated shared-memory tiling on 1,000 random GEMM shapes, showing a robust $1.15\times$ mean speedup over naive GPU kernels and 100–$500\times$ over a serial CPU baseline.

Undergraduate Researcher Jan 2024 – Jun 2025
Advanced Institute of Natural Sciences, Beijing Normal University, Zhuhai *Supervisor: Prof. Jianxiong Guo*

- Co-first-authored a WASA 2025 paper [4] on clustered federated learning for IoT: designed a dynamic spectral-clustering scheme that groups clients by data heterogeneity to improve accuracy and convergence while preserving privacy.
- Implemented the adaptive clustering algorithms and a dynamic resource-allocation strategy in Python and evaluated scalability on large-scale simulated IoT networks.

Research Assistant Jan 2024 – Jun 2024
Malicious Traffic Detection Project, Nanjing Normal University (remote) *Supervisor: Dr. Nian Xia*

- Collected and preprocessed network-traffic datasets (TII-SSRC-23, UNSW-NB15), trained SVM and decision-tree detectors with the EFS-CST feature-selection method, and wrote the data-collection and model-evaluation sections of the AsiaJCIS 2024 paper [5].

Industry Experience

iFLYTEK Co., Ltd. Jul 2026 – Aug 2026
AI Infra Intern *Hefei, China*

- Developed and debugged an automated evaluation platform for adapting speech models (ASR/TTS) to heterogeneous domestic AI accelerators (Ascend NPU, DCU, MLU, MUSA); resolved 30 of 47 triaged engineering issues, redesigned task-liveness detection, and built a unified result-integrity validator that flagged all 99 historical faulty records with zero false positives in replay.
- Root-caused a WER degradation of Parakeet-CTC-1.1B on Ascend NPU via layer-wise operator comparison (1,811 ops): an NPU-only branch in HuggingFace Transformers cast the float relative-position-bias mask to boolean, discarding positional information across 42 attention layers; fix verified on a 200-sample regression and reported upstream.
- Led a cross-platform accuracy-drift audit against GPU baselines, separating engineering artifacts from genuine platform drift across 8 outlier models and establishing comparability pre-checks (model ID, weights, inference engine, decoding parameters) for chip-accuracy statistics.

Shenzhen Quant-Cloud Energy Network Technology Co., Ltd. Aug 2024
R&D Intern, Artificial Intelligence Department *Shenzhen, China*

- Built a bearing-temperature anomaly detector for wind-turbine gearboxes using Anomaly Transformer on historical sensor data for predictive maintenance; optimized preprocessing to improve detection accuracy.

Kenexs Intelligent Technology Co., Ltd. Jul 2023 – Sep 2023
Software Development Intern *Shenzhen, China*

- Built a Qt-based log analyzer to visualize and filter logs from industrial machine-vision cameras, improving testers' debugging efficiency; fixed critical bugs and refined the interface for robustness.

Skills

Programming: Python, C/C++, CUDA, Java, SQL, JavaScript
ML & Systems: PyTorch, HuggingFace Transformers, SGLang, NeMo, NumPy/SciPy, CUDA kernel optimization (Tensor Cores, cuBLAS), Linux, Git, LaTeX
Languages: Chinese (native), English (TOEFL iBT 94, Nov 2024)